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CHAPTER

Chemical Kinetics

CONTENTS

So far we have studied equilibrium reactions. In these reactions, the rates of the two opposing reactions are equal and the concentrations of reactants or products do not change with lapse of time. But most chemical reactions are **spontaneous reactions.** These reactions occur from left to right till all the reactants are converted to products. A spontaneous reaction may be slow or it may be fast. For example, the reactions between aqueous sodium chloride and silver nitrate is a fast reaction. The precipitate of $AgCl$ is formed as fast as $AgNO_3$ solution is added to $NaCl$ solution. On the contrary, the rusting of iron is a slow reaction that occurs over the years.

The branch of Physical chemistry which deals with the rate of reactions is called Chemical Kinetics. The study of Chemical Kinetics includes :

(1) The rate of the reactions and rate laws.

(2) The factors as temperature, pressure, concentration and catalyst, that influence the rate of a reaction.

(3) The mechanism or the sequence of steps by which a reaction occurs.

The knowledge of the rate of reactions is very valuable to understand the chemical of reactions. It is also of great importance in selecting optimum conditions for an industrial process so that it proceeds at a rate to give maximum yield.

REACTION RATE

The rate of a reaction tells as to what speed the reaction occurs. Let us consider a simple reaction

$$A \longrightarrow B$$

The concentration of the reactant A decreases and that of B increases as time passes. **The rate of reactions is defined as the change in concentration of any of reactant or products per unit time**. For the given reaction the rate of reaction may be equal to the rate of disappearance of A which is equal to the rate of appearance of B.

Thus

$$\text{rate of reaction} = \text{rate of disappearance of } A$$
$$= \text{rate of appearance of } B$$

or

$$\text{rate} = -\frac{d[A]}{dt}$$
$$= +\frac{d[B]}{dt}$$

where [ ] represents the concentration in moles per litre whereas 'd' represents infinitesimally small change in concentration. Negative sign shows the concentration of the reactant A decreases whereas the positive sign indicates the increase in concentration of the product B.

UNITS OF RATE

Reactions rate has the units of concentration divided by time. We express concentrations in moles per litre (mol/litre or mol/l or mol l^{-1}) but time may be given in any convenient unit second (s), minutes (min), hours (h), days (d) or possible years. Therefore, the units of reaction rates may be

mole/litre sec	or	mol l^{-1} s
mole/litre min	or	mol l^{-1} min^{-1}
mole/litre hour	or	mol l^{-1} h^{-1} and, so on

RATE LAWS

At a fixed temperature the rate of a given reaction depends on concentration of reactants. The exact relation between concentration and rate is determined by measuring the reaction rate with different initial reactant concentrations. By a study of numerous reactions it is shown that : **the rate of a reaction is directly proportional to the reactant concentrations, each concentration being raised to some power.**

Thus for a substance A undergoing reaction,

rate $\quad$ rate $\propto [A]^n$

or $\quad$ rate $= k\,[A]^n$ $\quad$ ...(1)

For a reaction

$$2A + B \longrightarrow \text{products}$$

the reaction rate with respect to A or B is determined by varying the concentration of one reactant, keeping that of the other constant. Thus the rate of reaction may be expressed as

$$\text{rate} = k\,[A]^m\,[B]^n \qquad ...(2)$$

Expressions such as (1) and (2) tell the relation between the rate of a reaction and reactant concentrations.

An expression which shows how the reaction rate is related to concentrations is called the rate law or rate equation.

The power (exponent) of concentration n or m in the rate law is usually a small whole number integer (1, 2, 3) or fractional. The proportionality constant k is called the **rate constant** for the reaction.

Examples of rate law :

	REACTIONS	RATE LAW
(1)	$2N_2O_5 \longrightarrow 4NO_2 + O_2$	rate $= k\,[N_2O_5]$
(2)	$H_2 + I_2 \longrightarrow 2HI$	rate $= k\,[H_2]\,[I_2]$
(3)	$2NO_2 \longrightarrow 2NO + O_2$	rate $= k\,[NO_2]^2$
(4)	$2NO + 2H_2 \longrightarrow N_2 + 2H_2O$	rate $= k\,[H_2]\,[NO]^2$

In these rate laws where the quotient or concentration is not shown, it is understood to be 1. That is $[H_2]^1 = [H_2]$.

It is apparent that the rate law for a reaction must be determined by experiment. It cannot be written by merely looking at the equation with a background of our knowledge of Law of Mass Action. However, for some elementary reactions the powers in the rate law may correspond to coefficients in the chemical equation. **But usually the powers of concentration in the rate law are different from coefficients.** Thus for the reaction (4) above, the rate is found to be proportional to $[H_2]$ although the quotient of H_2 in the equation is 2. For NO the rate is proportional to $[NO]^2$ and power '2' corresponds to the coefficient.

ORDER OF A REACTION

The order of a reaction is defined as the sum of the powers of concentrations in the rate law.

Let us consider the example of a reaction which has the rate law

$$\text{rate} = k\,[A]^m\,[B]^n \qquad ...(1)$$

The order of such a reaction is $(m + n)$.

The order of a reaction can also be defined with respect to a single reactant. Thus the reaction order with respect to A is m and with respect to B it is n. The **overall order of reaction** $(m + n)$ may range from 1 to 3 and can be fractional.

Examples of reaction order :

RATE LAW	REACTION ORDER
rate = $k\,[N_2O_5]$	1
rate = $k\,[H_2]\,[I_2]$	$1 + 1 = 2$
rate = $k\,[NO_2]^2$	2
rate = $k\,[[H_2]\,[NO]^2$	$1 + 2 = 3$
rate = $k\,[CHCl_3]\,[Cl_2]^{1/2}$	$1 + \frac{1}{2} = 1\frac{1}{2}$

Reactions may be classified according to the order. If in the rate law (1) above

$m + n = 1$, it is **first order reaction**

$m + n = 2$, it is **second order reaction**

$m + n = 3$, it is **third order reaction**

ZERO ORDER REACTION

A reactant whose concentration does not affect the reaction rate is not included in the rate law. In effect, the concentration of such a reactant has the power 0. Thus $[A]^0 = 1$.

A zero order reaction is one whose rate is independent of concentration. For example, the rate law for the reaction

$$NO_2 + CO \longrightarrow NO + CO_2$$

at 200° C is

$$\text{rate} = k\,[NO_2]^2$$

Here the rate does not depend on [CO], so this is not included in the rate law and the power of [CO] is understood to be zero. The reaction is **zeroth order** with respect to CO. The reaction is second order with respect to $[NO_2]$. The overall reaction order is $2 + 0 = 2$.

MOLECULARITY OF A REACTION

Chemical reactions may be classed into two types :

(a) Elementary reactions

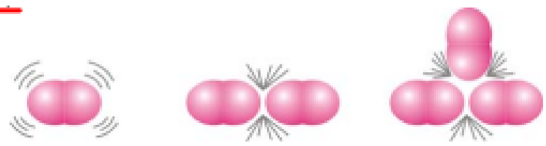
(b) Complex reactions

An **elementary reaction** is a simple reaction which occurs in a single step.

A **complex reaction** is that which occurs in two or more steps.

Molecularity of an Elementary Reaction

The molecularity of an elementary reaction is defined as : **the number of reactant molecules involved in a reaction.**



Unimolecular Bimolecular Termolecular

Differences Between Order and Molecularity

Order of a Reaction	Molecularity of a Reaction
1. It is the **sum of powers** of the concentration terms in the rate law expression.	1. It is number of **reacting species** undergoing simultaneous collision in the elementary or simple reaction.
2. It is an **experimentally** determined value.	2. It is a **theoretical** concept.
3. It can have **fractional** value.	3. It is always a **whole number.**
4. It can assume **zero value.**	4. It **can not have zero value.**
5. Order of a reaction **can change** with the conditions such as pressure, temperature, concentration.	5. Molecularity is **invariant** for a chemical equation.

PSEUDO-ORDER REACTIONS

A reaction in which one of the reactants is present in a large excess shows an order different from the actual order. **The experimental order which is not the actual one is referred to as the pseudo order.** Since for elementary reactions molecularity and order are identical, pseudo-order reactions may also be called **pseudo molecular reactions.**

Let us consider a reaction

$$A + B \longrightarrow \text{products}$$

in which the reactant B is present in a large excess. Since it is an elementary reaction, its rate law can be written as

$$\text{rate} = k[A][B]$$

As B is present in large excess, its concentration remains practically constant in the course of reaction. Thus the rate law can be written as

$$\text{rate} = k'[A]$$

where the new rate constant $k' = k[B]$. Thus the actual order of the reaction is second-order but in practice it will be first-order. Therefore, the reaction is said to have a **pseudo-first order.**

Examples of Pseudo-order Reactions

(1) **Hydrolysis of an ester.** For example, ethyl acetate upon hydrolysis in aqueous solution using a mineral acid as catalyst forms acetic acid and ethyl alcohol.

$$\underset{\text{ethyl acetate}}{CH_3COOC_2H_5} + \underset{\text{(excess)}}{H_2O} \longrightarrow \underset{\text{acetic acid}}{CH_3COOH} + \underset{\text{ethyl alcohol}}{C_2H_5OH}$$

Here a large excess of water is used and the rate law can be written as

$$\begin{aligned} \text{rate} &= k[CH_3COOH][H_2O] \\ &= k'[CH_3COOH] \end{aligned}$$

The reaction is actually second-order but in practice it is found to be first-order. Thus it is a pseudo-first order reaction.

(2) **Hydrolysis of sucrose.** Sucrose upon hydrolysis in the presence of a dilute mineral acid gives glucose and fructose.

$$\underset{\text{sucrose}}{C_{12}H_{22}O_{11}} + \underset{\text{(excess)}}{H_2O} \longrightarrow \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

If a large excess of water is present, $[H_2O]$ is practically constant and the rate law may be written as

$$\text{rate} = k\,[C_{12}H_{22}O_{11}]\,[H_2O]$$
$$= k\,[C_{12}H_{22}O_{11}]$$

The reaction though of second-order is experimentally found to be first-order. Thus it is a pseudo-first-order reaction.

ZERO ORDER REACTIONS

In a zero order reaction, rate is independent of the concentration of the reactions. Let us consider a zero-order reaction of the type

	A	$\longrightarrow$ Products
Initial conc.	a	0
Final conc.	$a-x$	x

$$\text{Rate of reaction} = \frac{-d[A]}{dt} = k_0\,[A]^\circ$$

or
$$\frac{dx}{dt} = \frac{-d\,(a-x)}{dt} = k_0\,(a-x)^\circ = k_0$$

On integrating we get

$$\boxed{k_0 = \frac{x}{t} \quad \text{or} \quad x = k_0 t}$$

where k_0 is the rate constant of a zero-order reaction, the unit of which is concentration per unit time. In zero order reaction, the **rate constant is equal to the rate of reaction at all concentrations.**

FIRST ORDER REACTIONS

Let us consider a first order reaction

$$A \longrightarrow \text{products}$$

Suppose that at the beginning of the reaction ($t = 0$), the concentration of A is a moles litre^{-1}. If after time t, x moles of A have changed, the concentration of A is $a - x$. We know that for a first order reaction, the rate of reaction, dx/dt, is directly proportional to the concentration of the reactant. Thus,

$$\frac{dx}{dt} = k\,(a-x)$$

or
$$\frac{dx}{a-x} = k\,dt \qquad ...(1)$$

Integration of the expression (1) gives

$$\int \frac{dx}{a-x} = \int k\,dt$$

or
$$-\ln(a-x) = kt + \text{I} \qquad ...(2)$$

where I is the constant of integration. The constant k may be evaluated by putting $t = 0$ and $x = 0$. Thus,

$$\text{I} = -\ln a$$

Substituting for I in equation (2)

$$\ln\frac{a}{a-x} = kt \qquad ...(3)$$

or
$$k = \frac{1}{t}\ln\frac{a}{a-x}$$

Changing into common logarithms

$$k = \frac{2.303}{t} \log \frac{a}{a - x} \qquad ...(4)$$

The value of k can be found by substituting the values of a and $(a - x)$ determined experimentally at time interval t during the course of the reaction.

Sometimes the integrated rate law in the following form is also used :

$$k = \frac{2.303}{t_2 - t_1} \log \frac{(a - x_1)}{(a - x_2)}$$

where x_1 and x_2 are the amounts decomposed at time intervals t_1 and t_2 respectively from the start.

Examples of First order Reactions

Some common reactions which follow first order kinetics are listed below :

(1) **Decomposition of N_2O_5 in CCl_4 solution.** Nitrogen pentoxide in carbon tetrachloride solution decomposes to form oxygen gas,

$$N_2O_5 \longrightarrow 2NO_2 + \frac{1}{2}O_2$$

The reaction is carried in an apparatus shown in Fig. 20.4. The progress of the reaction is monitored by measuring the volume of oxygen evolved from time to time.

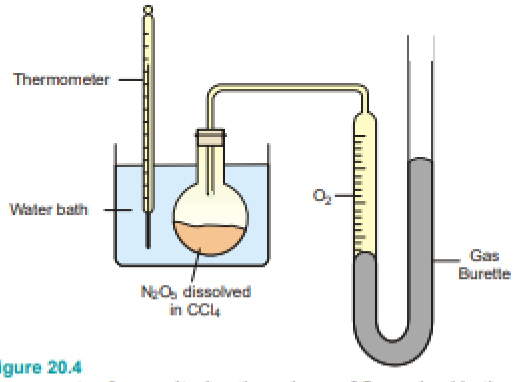


Figure 20.4
An apparatus for monitoring the volume of O_2 evolved in the decomposition of N_2O_5 dissolved in carbon tetrachloride.

If V_t be the volume of O_2 at any time t and V_∞ the final volume of oxygen when the reaction is completed, the V_∞ is a measure of the initial concentration of N_2O_5 and $(V_\infty - V_t)$ is a measure of undecomposed N_2O_5 $(a - x)$ remaining at time t. Thus,

$$k = \frac{2.303}{t} \log_{10} \frac{V_\infty}{V_\infty - V_t}$$

On substituting values of V_∞, $(V_\infty - V_t)$ at different time intervals, t, the value of k is found to be goes on changing since D-glucose rotates the plane of polarised light to the right and D-fructose to the left. **The change in rotation is proportional to the amount of sugar decomposed.**

Let the final rotation be r_∞, the initial rotation r_0 while the rotation at any time t is r_t

The initial concentration, a is $\propto (r_0 - r_\infty)$.

The concentration at time t, $(a - x)$ is $\propto (r_t - r_\infty)$

Substituting in the first order rate equation,

$$k = \frac{2.303}{t} \log_{10} \frac{a}{a-x}$$

we have

$$k = \frac{2.303}{t} \log_{10} \frac{(r_0 - r_\infty)}{(r_t - r_\infty)}$$

If the experimental values of t $(r_0 - r_\infty)$ and $(r_t - r_\infty)$ are substituted in the above equation, a constant value of k is obtained.

SOLVED PROBLEM. The optical rotation of sucrose in 0.9N HCl at various time intervals is given in the table below.

time (min)	0	7.18	18	27.1	∞
rotation (degree)	+24.09	+21.4	+17.7	+15	–10.74

Show that inversion of sucrose is a first order reaction.

SOLUTION

The available data is substituted in the first order rate equation for different time intervals.

$$k = \frac{2.303}{t} \log_{10} \frac{r_0 - r_\infty}{r_t - r_\infty}$$

$r_0 - r_\infty = 24.09 - (-10.74) = 34.83$ for all time intervals. Thus, the value of rate constant can be found.

time (t)	$r_t = r_\infty$	
		$k = \frac{1}{t} \log \frac{(r_0 - r_\infty)}{(r_t - r_\infty)}$
7.18	32.14	$k = \frac{1}{7.18} \log \frac{34.83}{32.14} = 0.0047$
18	28.44	$k = \frac{1}{18} \log \frac{34.83}{28.44} = 0.0048$
27.1	25.74	$k = \frac{1}{27.1} \log \frac{34.83}{25.74} = 0.0048$

Since the value of k comes out to be constant, the inversion of sucrose is a **first order reaction**.

SECOND ORDER REACTIONS

Let us take a second order reaction of the type

$$2A \longrightarrow \text{products}$$

Suppose the initial concentration of A is a moles litre^{-1}. If after time t, x moles of A have reacted, the concentration of A is $(a - x)$. We know that for such a second order reaction, rate of reaction is proportional to the square of the concentration of the reactant. Thus,

$$\frac{dx}{dt} = k(a-x)^2 \qquad ...(1)$$

where k is the rate constant, Rearranging equation (1), we have

$$\frac{dx}{(a-x)^2} = k\,dt \qquad ...(2)$$

On integration, it gives

$$\frac{1}{a-x} = kt + I \qquad ...(3)$$

where I is is integration constant. I can be evaluated by putting $x = 0$ and $t = 0$. Thus,

$$I = \frac{1}{a} \qquad ...(4)$$

Substituting for I in equation (3)

$$\frac{1}{a-x} = kt + \frac{1}{a}$$

$$kt = \frac{1}{a-x} - \frac{1}{a}$$

Thus

$$\boxed{k = \frac{1}{t} \cdot \frac{x}{a(a-x)}}$$

This is the integrated rate equation for a second order reaction.

Examples of Second order Reaction

THIRD ORDER REACTIONS

Let us consider a simple third order reaction of the type

$$3A \longrightarrow \text{products}$$

Let the initial concentration of A be a moles litre^{-1} and after time t, x, moles have reacted. Therefore, the concentration of A becomes $(a - x)$. The rate law may be written as :

$$\frac{dx}{dt} = k(a-x)^3 \qquad ...(1)$$

Rearranging equation (1), we have

$$\frac{dx}{(a-x)^3} = k\,dt \qquad ...(2)$$

On integration, it gives

$$\frac{1}{2(a-x)^2} = kt + I \qquad ...(3)$$

where I is the integration constant. I can be evaluated by putting $x = 0$ and $t = 0$. Thus,

$$I = \frac{1}{2a^2}$$

By substituting the value of I in (3), we can write

$$kt = \frac{1}{2(a-x)^2} - \frac{1}{2a^2}$$

Therefore,

$$\boxed{k = \frac{1}{t}.\frac{x(2a-x)}{2a^2(a-x)^2}}$$

This is the integrated rate equation for a third order reaction.

Examples of Third order Reactions

There are not many reactions showing third order kinetics. A few of the known examples are :

(*i*) $2FeCl_3(aq) + SnCl_2(aq) \longrightarrow 2FeCl_2 + SnCl_4$

(*ii*) $2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$

(*iii*) $2NO(g) + Cl_2(g) \longrightarrow 2NOCl(g)$

UNITS OF RATE CONSTANT

The units of rate constant for different orders of reactions are different.

Units of Zero order Rate constant

For a zero order reaction, the rate constant k is given by the expression

$$k = \frac{d[\mathrm{A}]}{dt} = \frac{\mathrm{mol}}{\mathrm{litre}} \times \frac{1}{\mathrm{time}}$$

Thus the units of k are

$$\mathbf{mol\ l^{-1}\ time^{-1}}$$

Time may be given in seconds, minutes, days or years.

Units of First order Rate constant

The rate constant of a first order reaction is given by

$$k = \frac{2.303}{t} \log \frac{[\mathrm{A}]_0}{[\mathrm{A}]_t}$$

Thus the rate constant for the first order reaction is independent of the concentration. It has the unit

$$\mathbf{time^{-1}}$$

Units of Second order Rate constant

The rate constant for a second order reaction is expressed as

$$k = \frac{1}{t} \times \frac{x}{[\mathrm{A}]_0 ([\mathrm{A}]_0 - x)}$$

or

$$= \frac{\text{concentration}}{\text{concentration} \times \text{concentration}} \times \frac{1}{\text{time}}$$

$$= \frac{1}{\text{concentration}} \times \frac{1}{\text{time}}$$

$$= \frac{1}{\text{mole/litre}} \times \frac{1}{\text{time}}$$

$$= \mathrm{mol^{-1}\ l\ time^{-1}}$$

Thus the units for k for a second order reactions are

$$\mathbf{mol^{-1}\ l\ time^{-1}}$$

Units of Third order Rate constant

The rate constant for a third order reaction is

$$k = \frac{1}{t} \cdot \frac{x(2a - x)}{2a^2 (a - x)^2}$$

or

$$k = \frac{\text{concentration} \times \text{concentration}}{(\text{concentration})^2 \times (\text{concentration})^2} \times \frac{1}{\text{time}}$$

$$= \frac{1}{(\text{concentration})^2} \times \frac{1}{\text{time}}$$

$$= \frac{1}{(\text{mol/litre})^2} \times \frac{1}{\text{time}}$$

Thus the units of k for third order reaction are

$$\mathbf{mol^{-2}\ l^2\ time^{-1}}$$

HALF-LIFE OF A REACTION

Reaction rates can also be expressed in terms of **half-life** or **half-life period.** It is defined as : **the time required for the concentration of a reactant to decrease to half its initial value.**

Calculation of Half-life of a First order Reaction

The integrated rate equation (4) for a first order reaction can be stated as :

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

where $[A]_0$ is initial concentration and [A] is concentration at any time t. Half-life, $t_{1/2}$, is time when initial concentration reduces to $\frac{1}{2}$ *i.e.*,

$$[A] = \frac{1}{2}[A]_0$$

Substituting values in the integrated rate equation, we have

$$k = \frac{2.303}{t_{1/2}} \log \frac{[A]_0}{1/2[A]_0} = \frac{2.303}{t_{1/2}} \log 2$$

or $$t_{1/2} = \frac{2.303}{k} \log 2 = \frac{2.303}{k} \times 0.3010$$

or $$t_{1/2} = \frac{0.693}{k}$$

It is clear from this relation that :

(1) half-life for a first order reaction is **independent of the initial concentration.**

(2) it is **inversely proportional ot k,** the rate-constant.

SOLVED PROBLEM 1. Compound A decomposes to form B and C the reaction is first order. At 25°C the rate constant for the reaction is 0.450 s^{-1}. What is the half-life of A at 25°C ?

SOLUTION

We know that for a first order reaction, half-life $t_{1/2}$, is given by the expression

$$t_{1/2} = \frac{0.693}{k}$$

where k = rate constant

Substituting the value of k = 0.450 s^{-1}, we have

$$t_{1/2} = \frac{0.693}{0.450\,s^{-1}} = \mathbf{1.54\ s}$$

Thus half-life of the reaction $A \rightarrow B + C$ is 1.54 seconds.

SOLVED PROBLEM 2. The half-life of a substance in a first order reaction is 15 minutes. Calculate the rate constant.

SOLUTION

For a first order reaction

$$t_{1/2} = \frac{0.693}{k}$$

Putting $t_{1/2}$ = 15 min in the expression and solving for k, we have

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{15\text{ min}} = \mathbf{4.62 \times 10^{-2}\ min^{-1}}$$

SOLVED PROBLEM 3. For the reaction

$$2N_2O_5 \longrightarrow 4NO_2 + O_2$$

the rate is directly proportional to $[N_2O_5]$. At 45°C, 90% of the N_2O_5 reacts in 3600 seconds. Find the value of the rate constant k.

SOLUTION

Since rate is $\propto [N_2O_5]$ it is first order reaction. The integrated rate equation is

$$k = \frac{2.303}{t} \log \frac{[N_2O_5]_0}{[N_2O_5]}$$

When 90% of N_2O_5 has reacted, the initial concentration is reduced to $\frac{1}{10}$ That is,

$$[N_2O_5] = \frac{1}{10}[N_2O_5]_0$$

Substituting values in the rate equation,

$$k = \frac{2.303}{3600} \log \frac{[N_2O_5]_0}{\frac{1}{10}[N_2O_5]_0}$$

$$= \frac{2.303}{3600} \log 10 = \frac{2.303}{3600} \times 1$$

Thus $$k = \frac{2.303}{3600} = \mathbf{6.40 \times 10^{-4}\ s^{-1}}$$

SOLVED PROBLEM 4. The rate law for the decomposition of $N_2O_5\ (l)$ is : rate $= k\,[N_2O_5]$ where $k = 6.22 \times 10^{-4}$ sec^{-1}. Calculate half-life of $N_2O_5\ (l)$ and the number of seconds it will take for an initial concentration of $N_2O_5\ (l)$ of 0.100 M to drop to 0.0100 M.

SOLUTION

Calculation of half-life

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{6.22 \times 10^{-4}\ \text{sec}^{-1}} = 1.11 \times 10^{3}\ \text{sec}$$

Calculation of time in seconds for drop of $[N_2O_5]$ from 0.100 M to 0.0100 M

From first order integrated rate equation,

$$t = \frac{2.303}{t} \log \frac{[N_2O_5]_0}{[N_2O_5]_t}$$

or $$t = \frac{2.303}{k} \log \frac{[N_2O_5]_0}{[N_2O_5]_t}$$

Substituting values

$$t = \frac{2.303}{6.22 \times 10^{-4}} \log \frac{0.100}{0.0100}$$

$$= \frac{2.303}{6.22 \times 10^{-4}} \times 1$$

$$= \mathbf{3.70 \times 10^{3}\ sec}$$

SOLVED PROBLEM 5. For a certain first order reaction $t_{0.5}$ is 100 sec. How long will it take for the reaction to be completed 75% ?

SOLUTION

Calculation of k

For a first order reaction

$$t_{1/2} = \frac{0.693}{k}$$

or $$100 = \frac{0.693}{k}$$

$\therefore$ $$k = \frac{0.693}{100} = 0.00693 \text{ sec}^{-1}$$

Calculation of time for 75% completion of reaction

The integrated rate equation for a first order reaction is

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

or $$t = \frac{2.303}{k} \log \frac{[A]_0}{[A]}$$

When $\frac{3}{4}$ initial concentration has reacted, it is reduced to $\frac{1}{4}$

Substituting values in the rate equation

$$t_{3/4} = \frac{2.303}{0.00693} \log \frac{[A]_0}{\frac{1}{4}[A]_0}$$

$$= \frac{2.303}{0.00693} \log 4 = \textbf{200 sec}$$

SOLVED PROBLEM 6. A first order reaction is one-fifth completed in 40 minutes. Calculate the time required for its 100% completion.

SOLUTION

Calculation of k

For a first order reaction

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

After 40 mts, the initial concentration is reduced to $\frac{4}{5}$ That is,

$$[A] = \frac{4}{5}[A]_0$$

Substituting values in the equation above

$$k = \frac{2.303}{40} \log \frac{[A]_0}{\frac{4}{5}[A]_0}$$

or $$k = \frac{2.303}{40} \log 5 - \log 4 = 0.00558 \text{ mt}^{-1}$$

Calculation of time required for 100% completion

We know that for first order reaction

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

$$t = \frac{2.303}{k} \log \frac{[A]_0}{[A]}$$

If reaction is 100% complete in, say, t_1 time, we have, [A] = 0. Thus,

$$t_1 = \frac{2.303}{0.00558} \log \frac{[A]_0}{0} = \infty$$

SOLVED PROBLEM 7. 50% of a first order reaction is complete in 23 minutes. Calculate the time required to complete 90% of the reaction.

SOLUTION

Calculation of k

$$t_{0.5} = \frac{0.693}{k}$$

or
$$k = \frac{0.693}{t_{0.5}} = \frac{0.693}{23} = 0.0301304 \text{ mt}^{-1}$$

Calculation of time for 90% completion of the reaction

For first order reaction, integrated rate equation is

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]} \quad ...(1)$$

or
$$t = \frac{2.303}{k} \log \frac{[A]_0}{[A]} \quad ...(2)$$

When 90% of the initial concentration has reacted, 10% of it is left. That is.

$$[A] = \frac{1}{10}[A]_0$$

Substituting values in equation (2)

$$t = \frac{2.303}{0.0301304} \log \frac{[A]_0}{\frac{1}{10}[A]_0} = \frac{2.303}{0.0301304} \log 10$$

$$= \frac{2.303}{0.0301304} = \mathbf{76.4\ min}$$

Half-life for a Second order Reaction

For the simple second order reaction 2A → Products, the integrated rate equation is

$$k\,t = \frac{1}{[A]} - \frac{1}{[A]_0}$$

where $[A]_0$ is the initial concentration and [A] is the concentration when time t has elapsed.

When one-half life has elapsed.

$$[A] = \frac{1}{2}[A]_0$$

and we have

$$k\,t_{1/2} = \frac{1}{\frac{1}{2}[A]_0} - \frac{1}{[A]_0}$$

or
$$k\,t_{1/2} = \frac{2}{[A]_0} - \frac{1}{[A]_0}$$

Solving for $t_{1/2}$ we find that

$$t_{1/2} = \frac{1}{k[A]_0}$$

(2) **Graphical method**

For reactions of the type A $\rightarrow$ products, we can determine the reaction order by seeing whether a graph of the data fits one of the integrated rate equations.

In case of First order

We have already derived the integrated rate equation for first order as

$$\ln \frac{a}{a-x} = k\,t$$

Simplifying, it becomes

$$\begin{array}{ccccc} \ln(a-x) & = & -kt & + & \ln a \\ \uparrow & & \uparrow & & \uparrow \\ y & = & mx & + & b \end{array}$$

Thus the two variables in the first order rate equation are :

$$\ln \frac{a}{a-x} \text{ and } t$$

Hence, if $\ln \frac{a}{a-x}$ is plotted against t and straight line results (Fig. 20.7), the corresponding reaction is of the first order. However, if a curve is obtained, the reaction is not first order.

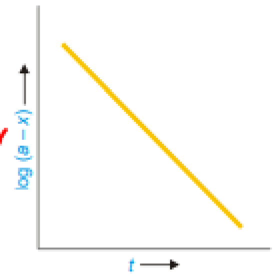


Figure 20.7
Plot of log(a – x) against t for a first order reaction.

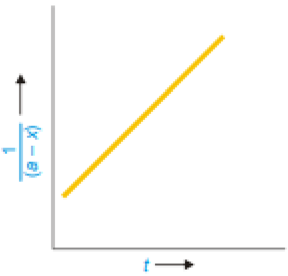


Figure 20.8
Plot of 1/(a – x) against t for a second order reaction.

In case of Second order

We have already shown that second order rate equation can be written as

$$\begin{array}{ccccc} \frac{1}{a-x} & = & kt & + & \frac{1}{a} \\ \uparrow & & \uparrow & & \uparrow \\ y & = & mx & + & b \end{array}$$

This is the equation of a straight line, $y = mx + b$. Here the two variables are

$$\frac{1}{a-x} \text{ and } t$$